Information Loss is Local: The Gravitational Refractive Index as Petz Recovery Fidelity

Sheng-Kai Huang[1,*]

[1] *Independent Researcher*

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The temporal asymmetry parameter $\tau = 1 - F \in [0, 1]$, established in Paper I from the Petz recovery framework, quantifies the arrow of time with the universal bound $F \geq e^{-\Sigma/2}$. Defining the gravitational entropy production $\Sigma_{\rm grav} \equiv -\ln(-g_{00}) = r_{\rm s}/r$, we obtain the identification $c_{\rm eff}/c = e^{-\Sigma_{\rm grav}/2} = F_{\rm bound}$: the coordinate speed of light *is* the Petz recovery fidelity bound [NEW SYNTHESIS]. We then observe that $\tau(r) = 1 - \sqrt{-g_{00}(r)}$ depends only on the *local* metric component g_{00}—no knowledge of the global causal structure is needed. This locality has a notable consequence: the operationally relevant boundary where $\tau = 1$ is the *apparent horizon*, not the event horizon. Using the Kodama vector for dynamical spacetimes, we show that τ tracks the apparent horizon in Vaidya collapse and the Hubble radius in FRW cosmology—consistent with the quantum extremal surface prescription. Since $\tau < 1$ everywhere for finite $\Sigma_{\rm grav}$, event horizons are excluded within this framework: information that falls in must eventually come out. The exponential metric $g_{00} = -e^{-r_{\rm s}/r}$, which saturates the Petz bound, predicts gravitational-wave echoes at $\Delta t \approx 4\, r_{\rm s}/c$—in principle testable with existing LIGO–Virgo data, though not yet searched for with appropriate templates.

INTRODUCTION

In Paper I [1], we established that the Petz recovery map [3]—the unique Bayesian retrodiction functor [6]—defines a temporal asymmetry parameter $\tau \equiv 1 - F(\rho, \mathcal{R}_{\sigma,\mathcal{N}} \circ \mathcal{N}(\rho)) \in [0, 1]$, bounded by [4, 5]

$$\tau \;\leq\; 1 - e^{-\Sigma/2}\,, \tag{1}$$

where Σ is the entropy production of the channel $\mathcal{N}$. $\tau = 0$ for closed (unitary) evolution; $\tau > 0$ for open systems.

General relativity admits a well-known reformulation in which gravity acts as a medium with position-dependent refractive index [7, 8]: the coordinate speed of light is $c_{\rm eff}/c = e^{\Phi/c^2}$, where $\Phi < 0$ is the Newtonian potential. In this paper we show that

$$\frac{c_{\rm eff}}{c} \;=\; e^{\Phi/c^2} \;=\; e^{-\Sigma_{\rm grav}/2} \;=\; F_{\rm bound}\,. \tag{2}$$

Light slowing down in a gravitational field *is* information loss; the Petz recovery fidelity *is* the refractive index.

This identification has a consequence that goes beyond translation. The quantity $\tau(r) = 1 - \sqrt{-g_{00}(r)}$ is *local*: to evaluate the degree of information loss at a spacetime point, one needs only g_{00} at that point—not the global causal structure, not the future evolution of the spacetime. In general relativity, event horizons are defined teleologically: determining whether a point lies inside an event horizon requires tracing null geodesics to future null infinity, which presupposes knowledge of the spacetime's entire future [9]. But τ makes no reference to the future. The boundary where $\tau = 1$ is set by *local* data, and that boundary is the *apparent horizon*—a quasi-local, causal surface [42, 43].

The logical chain of this paper is:

1. $c_{\rm eff}/c = F_{\rm bound}$ (Sec. : triple identification);
2. τ is local (Sec. : information loss does not require the future);
3. $\tau < 1$ everywhere for finite $\Sigma_{\rm grav}$ (Sec. : no event horizon within this framework);
4. echoes at $\Delta t \approx 4\, r_{\rm s}/c$ (Sec. : testable consequence).

THE TRIPLE IDENTIFICATION

The coordinate speed of light

[KNOWN] In the post-Newtonian approximation of GR, the coordinate speed of light in a static gravitational field is [7, 8]

$$\frac{c_{\rm eff}}{c} = e^{\Phi/c^2}\,, \tag{3}$$

where $\Phi < 0$ is the Newtonian gravitational potential. This result, first anticipated by Einstein in 1911 [7] and placed on rigorous footing by Dicke [8], is exact for the exponential metric and agrees with the Schwarzschild metric through $O(r_{\rm s}/r)^2$.

Gravitational entropy production

[NEW DEFINITION] We define the dimensionless gravitational entropy production as

$$\boxed{\Sigma_{\rm grav} \;\equiv\; \frac{2|\Phi|}{c^2} \;=\; \frac{r_{\rm s}}{r}\,,} \tag{4}$$

where $r_s = 2GM/c^2$ is the Schwarzschild radius. This is an *informational* entropy production—the cost of retrodiction from infinity to radius r—not a thermodynamic entropy in the Clausius sense. The Clausius entropy production for a photon in Tolman thermal equilibrium vanishes identically [23], because $T(\Phi)\sqrt{-g_{00}} = T_\infty$. By contrast, $\Sigma_{\rm grav}$ quantifies the informational cost of climbing the gravitational potential, which persists even in equilibrium.

Three independent arguments converge on Eq. (4):

(a) Modular flow [KNOWN].—In the entanglement first-law regime of AQFT on Schwarzschild, the fractional quantum relative entropy (QRE) loss between radius r and infinity is $(D_{\rm in} - D_{\rm out})/D_{\rm in} = r_s/r$, exactly, via the Tolman blueshift and energy redshift [27].

(b) Gravitational Landauer principle [KNOWN].—The Tolman-modified cost of erasing one bit at radius r exceeds that at infinity by a factor $1/\sqrt{-g_{00}}$. The dimensionless excess erasure cost yields $\Sigma_{\rm grav} = -\ln(-g_{00})$ [28], which equals r_s/r for the exponential metric and $-\ln(1 - r_s/r) \approx r_s/r + O((r_s/r)^2)$ for Schwarzschild.

(c) Quantum channel model [KNOWN].—Modeling gravitational redshift as a pure-loss bosonic channel with intensity transmissivity $\eta = -g_{00}$, the QRE decrease is $\Sigma = -\ln\eta = r_s/r$ [29].

All three routes share the common mathematical core $\Sigma_{\rm grav} = -\ln(-g_{00})$.

The gravitational thermal attenuator channel

[NEW RESULT] The three routes above are not merely analogies—they are facets of a single rigorous construction.

Theorem 1 (Gravitational thermal attenuator)**.** *For a free scalar field mode of frequency ω propagating radially from r_1 to $r_2 > r_1$ in a static metric $ds^2 = -f(r)\,c^2\,dt^2 + f(r)^{-1}dr^2 + r^2 d\Omega^2$, in the geometric optics limit ($\omega^2 \gg V_{\rm eff}$):* (a) *there exists a CPTP thermal attenuator $\mathcal{N}_{r_1\to r_2}$ with transmissivity $\eta = f(r_1)/f(r_2)$;* (b) *the environment is thermal at the local Tolman temperature $T_{\rm loc}(r_1) = T_H/\sqrt{f(r_1)}$;* (c) *the entropy production satisfies*

$$\Sigma_{\rm grav}(r) = -\ln f(r) + O(\omega^2/T^2) = -\ln(-g_{00})\,; \qquad (5)$$

(d) *the Petz recovery fidelity obeys $F \geq e^{-\Sigma_{\rm grav}/2} = \sqrt{f(r)}$;* (e) *greybody corrections enter at $O((r_s/r)^2)$ [66].*

Proof sketch.—The WKB mode equation gives amplitude transmission $|t_\omega|^2 = f(r_1)/f(r_2)$: the energy ratio (gravitational redshift) contributes $\sqrt{f(r_1)/f(r_2)}$ and the time-dilation ratio of proper times contributes an identical factor [63]. The Tolman relation identifies the environment as thermal, yielding a Gaussian thermal attenuator $\mathcal{E}_{\eta,N_B}$ [64] with explicit Kraus operators on Fock space. In the high-temperature limit, $\Sigma = -\ln\eta + O(\omega^2/T^2)$ [65], establishing (c). Part (d) follows from the JRSWW bound [5]. The leading greybody correction is $|R_0|^2 \sim (\omega r_s)^2$, giving $|\Sigma_{\rm exact} - \Sigma_0| \leq \frac{1}{4}(r_s/r)^2$ (see Supplemental Material [61]). □

TABLE I. Gravitational entropy production $\Sigma_{\rm grav} = r_s/r$ and Petz recovery fidelity bound $F_{\rm bound} = e^{-\Sigma_{\rm grav}/2}$ for representative astrophysical objects. The last row indicates the limit of the weak-field approximation.

Object	r_s/r	$F_{\rm bound}$	$\tau_{\max}$
Earth surface	1.4×10^{-9}	0.99999999930	7.0×10^{-10}
Sun surface	4.2×10^{-6}	0.9999979	2.1×10^{-6}
Neutron star	0.4	0.819	0.181
Schwarzschild $r = r_s$	1	0.607	0.393

Three corollaries follow immediately: (i) at the horizon ($f \to 0$), $\Sigma_{\rm grav} \to \infty$, $F \to 0$, $\tau \to 1$—complete information loss; (ii) in flat spacetime ($f = 1$), $\Sigma_{\rm grav} = 0$, $F = 1$, $\tau = 0$—no gravitational arrow of time; (iii) channels compose: $\Sigma_{\rm grav}(r_1 \to r_3) = \Sigma_{\rm grav}(r_1 \to r_2) + \Sigma_{\rm grav}(r_2 \to r_3)$. The gravitational refractive index satisfies $n_{\rm grav} = e^{\Sigma_{\rm grav}/2} = 1/F_{\rm bound}$: *refraction is inverse recovery* [8].

This theorem places the τ framework on a constructive footing: τ is the recovery failure of a specific, constructible CPTP map with explicit Kraus operators.

The triple identification

[NEW SYNTHESIS] Substituting Eq. (4) into the Petz fidelity bound $F \geq e^{-\Sigma/2}$ from Paper I:

$$F_{\rm bound} = e^{-\Sigma_{\rm grav}/2} = e^{\Phi/c^2}\,. \qquad (6)$$

Comparing with Eq. (3), we obtain the central result:

$$\boxed{\frac{c_{\rm eff}}{c} \;=\; e^{-\Sigma_{\rm grav}/2} \;=\; F_{\rm bound}\,.} \qquad (7)$$

The effective speed of light *is* the Petz recovery fidelity bound. Equivalently, the gravitational refractive index is the inverse fidelity:

$$n_{\rm grav}(r) = \frac{c}{c_{\rm eff}} = e^{+\Sigma_{\rm grav}/2} = \frac{1}{F_{\rm bound}}\,. \qquad (8)$$

Table I provides numerical verification across four orders of magnitude in gravitational compactness.

Caveat.—The identification (7) is exact for the exponential metric $g_{00} = -e^{-r_s/r}$ and agrees with Schwarzschild through $O(r_s/r)$. At $O((r_s/r)^2)$, the two metrics differ: the Schwarzschild expansion gives $g_{00}^{\rm Schw} = -1 + r_s/r - r_s^2/(2r^2) + \cdots$, identical to the exponential expansion through this order. The first discrepancy appears at $O((r_s/r)^3)$. In the solar system ($r_s/r \sim 10^{-6}$), this cubic correction is of order 10^{-18}—far below current measurement capability [16].

INFORMATION LOSS IS LOCAL

τ depends only on local g_{00}

[NEW SYNTHESIS] The temporal asymmetry at a spacetime point p at radius r from a mass M is

$$\tau(r) = 1 - \sqrt{-g_{00}(r)}\,. \tag{9}$$

This is a *local, operational* quantity: an observer at r determines τ by comparing a local clock with a reference clock at infinity. No information about the future evolution of the spacetime, no knowledge of what happens at other radii, and no global causal analysis is required.

This stands in sharp contrast with the event horizon. In general relativity, the event horizon is defined as

$$\mathcal{H}^+ = \partial J^-(\mathcal{I}^+)\,, \tag{10}$$

the boundary of the causal past of future null infinity [9]. To determine whether a point lies inside the event horizon, one must know the *entire future* of the spacetime—including whether the black hole eventually evaporates, accretes more mass, or merges with another. The event horizon is a teleological concept: it is defined by where light rays *will eventually* reach, not by what is happening *now*.

The τ framework replaces this global, teleological criterion with a local, operational one. Information loss is not a verdict delivered by the future—it is a condition determined by the present.

The apparent horizon as $\tau = 1$ surface

[NEW SYNTHESIS] In general relativity, the *apparent horizon* is defined as the outermost marginally outer trapped surface (MOTS): the surface where the expansion of outgoing null geodesics vanishes, $\theta_+ = 0$ [42]. Unlike the event horizon, the apparent horizon is quasi-local: it depends only on the metric and its first derivatives on a single time-slice.

The τ framework naturally selects this surface. For the Schwarzschild metric, $\tau = 1$ when $g_{00} = 0$, which occurs at $r = r_{\rm s}$—both the event horizon and the apparent horizon coincide. But in dynamical spacetimes they differ, and τ tracks the apparent horizon. This has a precise mathematical content:

$$\tau = 1 \iff g_{00} = 0 \iff \theta_+ = 0\ (\text{MOTS})\,. \tag{11}$$

The equivalence on the right holds for spherically symmetric spacetimes where $\theta_+ = (2/r)(1 - 2M(v)/r)^{1/2}$: $\theta_+ = 0$ precisely when $r = 2M(v)$, i.e., when $g_{00}^{\rm eff} = 0$.

Connection to QES.—[NEW SYNTHESIS] The quantum extremal surface (QES) prescription of Engelhardt and Wall [44] and its applications by Penington [32] and the island formula [33] identify the QES—not the classical event horizon—as the surface that controls the entanglement entropy of Hawking radiation. The QES extremizes the *generalized* entropy $S_{\rm gen} = A/(4G\hbar) + S_{\rm bulk}$ on a time-slice. The τ framework provides a complementary characterization: the QES is the surface where the *operational* information loss saturates ($\tau \to 1$), while the generalized entropy formulation controls the *counting* of degrees of freedom. Both agree that the operationally relevant boundary is quasi-local and causal—defined by data on a single slice, not by the global causal structure. In the semiclassical limit, the QES reduces to the apparent horizon [44], and $\tau = 1$ selects the same surface—the two prescriptions coincide.

Dynamic extension: the Kodama vector

[NEW SYNTHESIS] The static formulation of τ requires a timelike Killing vector ξ^a. The natural dynamical extension uses the Kodama vector [45]—the unique divergence-free timelike vector field in spherically symmetric spacetimes:

$$K^a = \epsilon^{ab}\nabla_b R\,, \tag{12}$$

where R is the areal radius and ϵ^{ab} is the Levi-Civita tensor on the (t, r) submanifold. In the static limit, K^a reduces to the Killing vector ξ^a, and $|K| = \sqrt{-g_{00}}$. The Kodama recovery fidelity is

$$F^K_{\rm bound}(A \to B) = \frac{|K|_A}{|K|_B}\,, \tag{13}$$

recovering all results of Sec. in the static limit.

Vaidya collapse.—[NEW SYNTHESIS] The ingoing Vaidya metric describes a spherically symmetric spacetime with a time-dependent mass function $M(v)$:

$$ds^2 = -\left(1 - \frac{2M(v)}{r}\right) dv^2 + 2\,dv\,dr + r^2\,d\Omega^2\,, \tag{14}$$

where v is the advanced (Eddington–Finkelstein) time and $dM/dv > 0$ for accretion [62]. The Kodama vector norm is [42, 46] $|K| = \sqrt{1 - 2M(v)/r}$, giving

$$\tau_K(r, v) = 1 - \sqrt{1 - \frac{2M(v)}{r}}\,. \tag{15}$$

This reaches $\tau = 1$ when $r = 2M(v)$—precisely the *apparent* horizon, the outermost MOTS on each $v = \text{const}$ slice. During gravitational collapse, $M(v)$ increases and the $\tau = 1$ surface expands outward, tracking the growing apparent horizon. During Hawking evaporation, $M(v)$ decreases and the $\tau = 1$ surface shrinks—information recovery improves, consistent with the Page curve.

Explicit demonstration: AH $\neq$ EH.—The event horizon is the outermost outgoing null geodesic that fails

TABLE II. τ_K at the event horizon versus the apparent horizon for the self-similar Vaidya spacetime $M(v) = \lambda v$. $\tau_K(\mathrm{AH}) = 1$ exactly for all λ; $\tau_K(\mathrm{EH}) < 1$. See Supplemental Material for derivations.

λ	$r_{\mathrm{EH}}/r_{\mathrm{AH}}$	$\tau_K(r_{\mathrm{EH}})$	$1-\tau_K(r_{\mathrm{EH}})$
0.001	1.004	0.937	6.3%
0.01	1.044	0.796	20.4%
0.03	1.162	0.627	37.3%
0.05	1.382	0.474	52.6%
1/16	2.000	0.293	70.7%

to escape to infinity. For radial null rays in Vaidya, $dr/dv = \frac{1}{2}(1 - 2M(v)/r)$, integrated backward from $r_{\mathrm{EH}} = 2M_0$ at $v \to \infty$. For the self-similar profile $M(v) = \lambda v$, the null geodesic equation admits the exact solution $r_{\mathrm{EH}}(v) = C_- v$, where $C_- = (1 - \sqrt{1 - 16\lambda})/4$, while $r_{\mathrm{AH}}(v) = 2\lambda v$. Since $C_- > 2\lambda$ for $\lambda > 0$ (see Supplemental Material [61] for proof), $r_{\mathrm{EH}} > r_{\mathrm{AH}}$ during accretion: the event horizon lies *outside* the apparent horizon.

Table II quantifies the discrepancy. At the event horizon, $\tau_K < 1$—the τ framework says the EH is *not* a surface of complete information loss. At the apparent horizon, $\tau_K = 1$ exactly, for all v. During evaporation ($dM/dv < 0$), the situation reverses: $r_{\mathrm{EH}} < r_{\mathrm{AH}}$, and the Kodama vector becomes spacelike at the EH. In both cases, $\tau_K = 1$ at the AH—never at the EH.

For a step-function collapse ($M(v) = M_0\,\theta(v)$), the event horizon originates at $r = 0$ when $v = -4M_0$—in the *flat* region, before the shell arrives. The EH "anticipates" the collapse; the AH appears only at $v = 0$. This illustrates the distinction clearly: τ_K is zero in the flat region (no information loss), even though the event horizon is already present.

FRW cosmology.—In a spatially flat FRW universe with Hubble parameter H,

$$\tau_K(R) = 1 - \sqrt{1 - H^2 R^2}\,, \tag{16}$$

reaching $\tau = 1$ at $R = 1/H$, the Hubble radius—which is the apparent horizon of FRW, not an event horizon [46].

Verification proposal: SYK model

[SPECULATIVE] The SYK model [49, 50] is a solvable quantum-mechanical model with a holographic dual to nearly-AdS_2 (JT) gravity, exhibiting both scrambling and black hole formation. It provides a setting where both sides of the τ equation—the QI recovery fidelity $F(t)$ and the gravitational entropy production $\Sigma_{\mathrm{grav}}(t)$—can be computed independently.

We propose the following test. In the SYK model with time-dependent coupling (modeling dynamical black hole formation and evaporation): (i) compute the Petz recovery fidelity $F(t)$ for an early-time operator from late-time data; (ii) compute $\Sigma_{\mathrm{grav}}(t) = -\ln(-g_{00}^{\mathrm{bulk}}(t))$ from the dual JT gravity; (iii) compare whether $F(t)$ tracks the apparent horizon or the event horizon in the bulk. The τ framework predicts that $F(t)$ tracks the apparent horizon. This calculation is feasible with exact diagonalization at moderate N [50]; the prediction awaits numerical verification.

NO EVENT HORIZON WITHIN THE τ FRAMEWORK

Model-independent no-horizon theorem

[NEW SYNTHESIS] The locality of τ established in Sec. has a direct physical consequence. If gravitational redshift is a quantum channel with *finite* entropy production $\Sigma < \infty$, then the Petz bound gives $F \geq e^{-\Sigma/2} > 0$, so $\tau < 1$. An event horizon requires $F = 0$ (complete information loss), which demands $\Sigma \to \infty$—impossible for a finite-mass object at finite distance. Under this premise, the Petz bound [5] excludes event horizons; neither the specific form of Σ_{grav} nor the saturation hypothesis is needed.

In the language of Sec. : because τ is local and finite, information loss is never complete. Within this framework, what falls in must eventually come out.

The exponential metric

[NEW SYNTHESIS] If the Petz bound is taken as an identity—$\sqrt{-g_{00}} = e^{-\Sigma_{\mathrm{grav}}/2}$—the metric component is uniquely determined. This gives the exponential metric [10, 11]:

$$ds^2 = -e^{-r_{\mathrm{s}}/r}\,c^2 dt^2 + e^{+r_{\mathrm{s}}/r}\big(dr^2 + r^2 d\Omega^2\big)\,, \tag{17}$$

in isotropic coordinates. Since $e^{-r_{\mathrm{s}}/r} > 0$ for all $r > 0$, $\tau < 1$ everywhere.

Exact solution.—The exponential metric is not a vacuum solution of the Einstein equations ($R_{\mu\nu} \neq 0$). It is an exact solution of Einstein gravity coupled to a phantom scalar field $\phi = M/r$ [17], and also arises in logarithmic $f(R)$ gravity [18] and scalar-Einstein-Gauss-Bonnet theory [19]. Makukov and Mychelkin [12] showed that three independent scalar-field families within GR all reduce to the exponential metric when the scalar charge equals the gravitational mass. Boonserm et al. [13] classified the geometry as a traversable wormhole.

The spatial metric is assumed, not derived.—The information-theoretic framework constrains only the temporal component g_{00} through $\sqrt{-g_{00}} = e^{-\Sigma_{\mathrm{grav}}/2}$. The spatial component $g_{rr} = e^{+r_{\mathrm{s}}/r}$ in Eq. (17) follows

from the isometry condition $g_{00} \cdot g_{rr} = -1$ (isotropic coordinates), which is equivalent to Dicke's scalar refractive index [8]: $n^2 = 1/(-g_{00})$ with $g_{rr} = n^2$. This is a physically motivated assumption—not a derivation from quantum information theory. The fundamental reason is that quantum channels model *temporal* processes (evolution, decoherence, loss), and the Petz recovery framework naturally constrains the temporal metric component; no analogous information-degradation process exists for the spatial direction. Consequently, we classify our predictions as *robust* (depending only on g_{00}: no-horizon theorem, refractive index, gravitational decoherence rate) or *conditional* (depending on the full metric: shadow size, ISCO radius, QNM frequencies, echo delay). The Jacobson [39] and Dorau–Much [27] programs *do* derive all metric components from information-theoretic premises, but their vacuum solution is Schwarzschild, not exponential. More recently, Thattarampilly et al. [41] solved the full Gravity-from-Entropy (GfE) field equations of Bianconi [40] for the static spherically symmetric vacuum, finding Schwarzschild with $O(r^{-4})$ corrections—again not the exponential metric. Every known entanglement-based gravity program thus produces Schwarzschild (plus perturbative corrections) as its vacuum solution. This gap remains an open question: the exponential metric is best understood as a *Petz saturation ansatz*—the geometry defined by promoting $F = e^{-\Sigma_{\rm grav}/2}$ to an identity—rather than the output of any established entropy-to-gravity derivation.

Comparison with Schwarzschild.—A persistent misconception holds that the exponential metric "predicts half the observed light bending." This conflates the Yilmaz *field equations*—an alternative theory of gravity shown to be inconsistent by Misner [14]—with the exponential *metric* as a solution of standard GR coupled to a phantom scalar. Particles and photons follow standard geodesics of (17); only the geometry differs from Schwarzschild.

The PPN expansion in isotropic coordinates gives

$$g_{00}^{\rm exp} = -e^{-r_{\rm s}/r} = -(1 - 2U + 2U^2 - \cdots)\,, \tag{18}$$

$$g_{ij}^{\rm exp} = e^{+r_{\rm s}/r}\,\delta_{ij} = (1 + 2U + 2U^2 + \cdots)\,\delta_{ij}\,, \tag{19}$$

where $U = r_{\rm s}/(2r)$ is the Newtonian potential. Matching with the standard PPN form $g_{00} = -(1 - 2U + 2\beta U^2 + \cdots)$, $g_{ij} = (1 + 2\gamma U + \cdots)\delta_{ij}$ yields $\gamma = \beta = 1$—*identical to GR*. Mychelkin et al. [15] verified explicitly that the perihelion precession at first post-Newtonian order matches Schwarzschild: $\Delta\phi_{\rm exp} = \Delta\phi_{\rm Schw} = 6\pi M/p$. The first difference appears at 2PN, where the precession differs by $\sim 2.3 \times 10^{-7}$ arcsec/century for Mercury—four orders of magnitude below current precision. All solar-system tests are passed: light bending, Shapiro delay, perihelion precession, and the Cassini bound $|\gamma - 1| < 2.3 \times 10^{-5}$ [16]. *The exponential metric is observationally degenerate with Schwarzschild for every current and foreseeable weak-field test.*

Caveat on saturation.—The JRSWW bound [5] is structurally strict for $\Sigma > 0$: the proof via the Golden–Thompson inequality is an equality only when the relevant operators commute—the $\Sigma = 0$ case [3]. No known quantum channel saturates the bound for $\Sigma > 0$. The exponential metric should be understood as the geometry defined by the *identity* $\sqrt{-g_{00}} = e^{-\Sigma_{\rm grav}/2}$; whether nature saturates this bound or admits a gap is empirical.

STRONG-FIELD PREDICTIONS

[CONDITIONAL] In the strong-field regime, the exponential and Schwarzschild metrics diverge, yielding three testable predictions. All depend on the spatial metric assumption $g_{rr} = e^{+r_{\rm s}/r}$.

(1) Shadow size.—The exponential photon sphere at $r_{\rm ph} = r_{\rm s}$ (isotropic coordinates) yields a critical impact parameter $b_{\rm crit} = 2e\,m \approx 5.44\,m$ versus Schwarzschild's $3\sqrt{3}\,m \approx 5.20\,m$ ($m = GM/c^2$): a 4.6% larger shadow. The EHT measurement of 42 ± 3 μas for M87* [47] is consistent with both metrics. This prediction awaits next-generation VLBI observations with sufficient angular resolution to distinguish the two.

(2) Quasinormal modes.—In the eikonal limit, $f_{\rm QNM}^{\rm exp}/f_{\rm QNM}^{\rm Schw} = 3\sqrt{3}/(2e) \approx 0.956$, a $\sim 4.4\%$ downward shift. WKB calculations find a -7.8% shift for a generalized exponential wormhole [18]. This prediction has not yet been tested; next-generation gravitational-wave detectors (Einstein Telescope, Cosmic Explorer) are expected to reach $\sim 1\%$ precision in QNM frequencies.

(3) Gravitational-wave echoes [MOST DISCRIMINATING PREDICTION].—A concrete and falsifiable signature of the τ framework. The traversable wormhole structure of the exponential metric produces echoes—repeated, damped pulses following the initial ringdown. The echo delay is set by the round-trip light travel time in the tortoise coordinate $r^* = \int e^{r_{\rm s}/r}\,dr$. For the exponential metric, r^* remains bounded (no logarithmic divergence as in Schwarzschild), and the effective potential cavity extends from the photon sphere at $r_{\rm ph} = r_{\rm s}$ (isotropic) to the wormhole throat at $r_{\rm throat} = r_{\rm s}/2$. The echo delay is

$$\Delta t_{\rm echo} = \frac{2}{c}\int_{r_{\rm throat}}^{r_{\rm ph}} e^{r_{\rm s}/r}\,dr \;\approx\; \frac{4\,r_{\rm s}}{c}\ln\!\left(\frac{r_{\rm s}}{r_{\rm throat}}\right), \tag{20}$$

where the logarithmic approximation holds because $e^{r_{\rm s}/r}$ is sharply peaked near the throat. Numerically, the exact integral gives $\Delta t_{\rm echo} \approx 4.2\,r_{\rm s}/c$. Table III gives explicit estimates for astrophysical sources across the LIGO and LISA bands.

The 2.5 ms echo for stellar-mass mergers falls in the LIGO–Virgo–KAGRA sensitivity band (20–2000 Hz; the echo frequency $\sim 1/\Delta t \sim 400$ Hz is near peak sensitivity). For supermassive black holes, the echo timescale

TABLE III. Echo delay $\Delta t_{\rm echo} \approx 4.2\, r_{\rm s}/c$ for representative sources compared with Planck-wall models [48] where $\Delta t_{\rm Planck} \approx r_{\rm s} \ln(r_{\rm s}/\ell_P)/c$. The exponential-metric echoes are ~ 2 orders of magnitude shorter.

Source	M	$\Delta t_{\rm echo}$	$\Delta t_{\rm Planck}$
Stellar BH (GW150914)	$60\, M_\odot$	2.5 ms	~ 230 ms
Intermediate-mass BH	$10^3\, M_\odot$	40 ms	~ 4 s
Sgr A*	$4{\times}10^6\, M_\odot$	165 s	$\sim 1.9{\times}10^4$ s
M87*	$6.5{\times}10^9\, M_\odot$	74 hr	~ 350 d

TABLE IV. Falsifiable predictions of the exponential metric (conditional on $g_{rr} = e^{+r_{\rm s}/r}$). None has been tested to date.

Observable	**Prediction**	**Schwarzschild**
Shadow $b_{\rm crit}/m$	$2e \approx 5.44$	$3\sqrt{3} \approx 5.20$
QNM shift	-4.4%	(reference)
Echo delay (stellar)	≈ 2.5 ms	none (horizon)

of minutes to hours is within the LISA band (10^{-4}–10^{-1} Hz). This is $\sim 90\times$ shorter than Planck-wall models [48]; current echo searches use ~ 200 ms templates appropriate for Planck-wall scenarios and would *miss* the exponential-metric signal. Dedicated short-delay searches in existing LIGO–Virgo O3/O4 data would be valuable.

The echo prediction follows directly from the no-horizon theorem: since $\tau < 1$ everywhere, the gravitational potential forms a well, not a one-way membrane. Gravitational waves enter, scatter off the wormhole throat, and return—carrying the signature of incomplete information loss. This echo prediction has not yet been tested with appropriate short-delay templates. A null detection with templates matched to Eq. (20) would falsify the saturation hypothesis ($\sqrt{-g_{00}} = e^{-\Sigma_{\rm grav}/2}$), while a positive detection would provide direct evidence for the exponential metric.

Summary of falsifiable predictions.—Table IV collects the three concrete predictions with numerical values. All three await experimental verification.

Honest assessment.—Since both metrics share $\gamma = \beta = 1$, weak-field data cannot distinguish them; the first differences are inherently *strong-field* effects. If nature selects $F > e^{-\Sigma_{\rm grav}/2}$, the actual metric lies between the exponential and Schwarzschild predictions. We emphasize a two-layer structure: the *robust* predictions (no-horizon theorem, $n_{\rm grav} = 1/F_{\rm bound}$, decoherence bound) depend only on g_{00} and hold regardless of the spatial metric; the *conditional* predictions (shadow, QNM, echoes) depend on the isometry assumption $g_{rr} = e^{+r_{\rm s}/r}$. The most characteristic signature—short-delay echoes (20)—has not yet been searched for with appropriate templates in LIGO–Virgo data.

SUPPORTING CONNECTIONS

The τ framework connects to several established results in gravitational physics. We classify each connection in Table V.

Gravitational decoherence.—[NEW SYNTHESIS] Pikovski et al. [22] showed that gravitational time dilation universally dephases composite quantum systems. In the τ framework [2],

$$\tau_{\rm grav}(t) = 1 - \exp\big(-N\Gamma t^2\big)\,, \tag{21}$$

where $\Gamma = (\Delta E)^2(\Delta\Phi)^2/(2\hbar^2 c^4)$. The dephasing rate is controlled by $\Delta\Phi/c^2 \sim \Sigma_{\rm grav}/2$: the gravitational entropy production governs both the static (refractive index) and dynamic (decoherence) aspects of information loss.

Unruh effect.—[NEW SYNTHESIS] A uniformly accelerating observer at acceleration a perceives a thermal bath at $T_U = \hbar a/(2\pi c k_{\rm B})$, inducing entropy production $\Sigma_{\rm Unruh} = 2\pi a \Delta x/(\hbar c^2)$ on any carried system [24, 25]. By the equivalence principle, this connects to $\Sigma_{\rm grav}$.

Bekenstein bound.—[NEW SYNTHESIS] The bound $S \le 2\pi E R/(\hbar c)$ [26] translates to $\tau \le 1 - \exp(-\pi E R/(\hbar c))$, providing a universal upper limit on temporal asymmetry for spatially bounded systems.

Complementarity.—[NEW SYNTHESIS] The infalling observer experiences $\tau_{\rm in} \approx 0$ (locally flat), while the external observer describes $\tau_{\rm ext} \to 1$ [30]. Different observer foliations assign different τ values—complementarity is observer-dependent τ.

Page curve.—[NEW SYNTHESIS] The Page curve [31] translates to $\tau(t) \le 1 - e^{-S_{\rm Page}(t)/2}$: τ grows before the Page time (information loss worsens) and shrinks after it (recovery improves) [32, 33].

Hayden–Preskill.—[KNOWN] Cotler et al. [35] proved that the HP decoder [34] *is* the Petz recovery map, with fidelity satisfying the same bound (1).

Entanglement wedge.—[KNOWN] In AdS/CFT, bulk reconstruction via the Petz map [36, 37] satisfies $F^2 \ge e^{-\Delta S_{\rm gen}}$ [35]—the same universal recovery inequality.

WHAT τ CAN AND CANNOT EXPLAIN

Can:

- Identify information loss as a local, operational quantity that does not require knowledge of the future.
- Identify the apparent horizon—not the event horizon—as the operationally relevant boundary, consistent with the QES prescription.
- Unify quantum error correction, thermodynamic irreversibility, black hole information, and gravitational decoherence under a single parameter τ.

TABLE V. Classification of results.

Connection	Status
$c_{\text{eff}}/c = F_{\text{bound}}$ (refractive index)	NEW SYNTHESIS
$\Sigma_{\text{grav}} = -\ln(-g_{00})$ (entropy production)	NEW DEFINITION
τ is local $\to$ apparent horizon	NEW SYNTHESIS
Kodama extension to dynamic spacetimes	NEW SYNTHESIS
No-horizon theorem	NEW SYNTHESIS
Observer-dependent τ (complementarity)	NEW SYNTHESIS
Page curve as $\tau(t)$	NEW SYNTHESIS
Gravitational decoherence as Petz recovery	NEW SYNTHESIS
HP decoder = Petz map [35]	KNOWN
Entanglement wedge via Petz [37]	KNOWN
$F^2 \geq e^{-\Delta S_{\text{gen}}}$ [32]	KNOWN
SYK verification of τ tracking AH	SPECULATIVE

- Provide universal bounds: any gravitational channel satisfies $F \geq e^{-\Sigma_{\text{grav}}/2}$, constraining recovery independently of the specific protocol.
- Derive the no-horizon theorem: $\tau < 1$ for any finite-mass object at finite distance (model-independent [5]).
- Extend to dynamical spacetimes via the Kodama vector, tracking apparent horizons in Vaidya and FRW.

Cannot:

- Explain dark matter: at galactic scales, $\Sigma_{\text{grav}} \sim 10^{-6}$, far too small to produce observable effects beyond Newtonian gravity.
- Explain dark energy or the cosmological constant.
- Derive g_{ab} from τ alone: the relation is one-directional ($g \to \Sigma \to \tau$); τ constrains only g_{00}. The spatial component g_{rr} follows from the Dicke isometry condition; programs that derive all components from entanglement equilibrium (Jacobson, Dorau–Much, Bianconi GfE [40]) yield Schwarzschild (plus perturbative corrections [41]) as the *vacuum* solution. The exponential metric requires a phantom scalar source, whose information-theoretic origin remains open.
- Provide the full non-perturbative Bogoliubov calculation beyond geometric optics: the channel of Theorem 1 is derived in the WKB limit, with greybody corrections bounded at $O((r_{\text{s}}/r)^2)$.
- Resolve the firewall paradox [38]: while $\tau < 1$ removes the event horizon premise, the framework does not address the interior structure of compact objects.

DISCUSSION AND OUTLOOK

The central result of this paper is that gravitational information loss is *local*. The temporal asymmetry $\tau = 1 - \sqrt{-g_{00}}$ depends only on the metric component at a single point. The operationally relevant boundary for information loss is therefore the apparent horizon—a quasi-local, causal surface [43]—not the event horizon. This is consistent with the quantum extremal surface prescription [32, 44] and the crossed-product algebra approach [58, 59], which makes gravitational entropy well-defined and observer-dependent [60].

The identification $c_{\text{eff}}/c = F_{\text{bound}}$ (Eq. (7)) is complementary to the Jacobson program [39]: while Jacobson derives *field equations* from thermodynamic entropy, we derive the *metric* directly from information recoverability. The Dorau–Much result [27]—deriving the semiclassical Einstein equations from quantum relative entropy—suggests that the two programs share a common informational foundation. Cirafici [56] has established fluctuation theorems in gravitational algebras, confirming that the Crooks–Petz connection extends to de Sitter spacetime.

We emphasize the logical structure of this paper. The results of Secs. – —the triple identification, locality of τ, and the no-horizon theorem—hold for *any* static metric satisfying $\Sigma_{\text{grav}} = -\ln(-g_{00})$ with $\Sigma_{\text{grav}} < \infty$; they are independent of whether the metric is Schwarzschild, exponential, or the Schwarzschild-plus-corrections solution of Gravity-from-Entropy [40, 41]. The exponential metric enters only in Sec. , where it serves as a maximal-recovery ansatz that saturates $F = e^{-\Sigma_{\text{grav}}/2}$. All currently known entanglement-based gravity programs (Jacobson, Dorau–Much, Bianconi GfE) produce Schwarzschild as the leading-order vacuum solution, with perturbative corrections [41]. Whether nature saturates the Petz bound or admits a gap of order $O((r_{\text{s}}/r)^2)$ is empirical—and the echo prediction of Sec. is precisely the observable that distinguishes the two possibilities.

Several directions remain. *(i) Beyond geometric optics*: the gravitational thermal attenuator (Theorem 1) is established in the WKB limit; computing the exact greybody spectrum for the exponential metric would determine whether the channel *saturates* the Petz bound or admits a gap of order $O((r_{\text{s}}/r)^2)$. *(ii) Rotation*: the Kerr analog of the exponential metric [20, 21] lacks both horizons and ergospheres; extending the framework beyond spherical symmetry requires the axisymmetric Kodama-type vector of Dorau and Verch [57]. *(iii) Entanglement-assisted recovery*: [NEW SYNTHESIS] When the gravitational channel $\mathcal{N}_{\text{grav}}$ acts on a system L that is entangled with a reference R, the total QRE decrease decomposes as $\Delta D_{\text{total}} = \Sigma_{\text{grav}} + I(L;R) - I(L';R) \geq 0$ (by the data-processing inequality), where $L' = \mathcal{N}(L)$. The single-party bound $F \geq e^{-\Sigma_{\text{grav}}/2}$ is derived without

access to R; when the recovery map has access to the quantum side information in R, the assisted fidelity satisfies $F_{\text{assisted}} \geq e^{-(\Sigma_{\text{grav}}+I(L;R)-I(L';R))/2}$. If the channel approximately preserves entanglement ($I(L';R) \approx I(L;R)$), then $F_{\text{assisted}} \gg e^{-\Sigma_{\text{grav}}/2}$: the recovery far exceeds the single-party geometric bound. This is precisely the mechanism behind the Gao–Jafferis–Wall traversable wormhole [51] and the Hayden–Preskill protocol [34]—shared entanglement provides a quantum side channel that enhances recovery without making the total entropy production negative. The assisted vs. unassisted gap $\Delta F \equiv F_{\text{assisted}} - e^{-\Sigma_{\text{grav}}/2}$ is a continuous measure that unifies the Page curve transition (growing $I(L';R)$) with wormhole traversability (high $I(L';R)/I(L;R)$). This prediction is testable via SYK numerics and quantum circuit experiments [52]. *(iv) Non-Markovian gravity*: if the gravitational channel exhibits memory effects, Σ_{grav} may not be additive and the Petz bound requires modification [24]. This could serve as a signature of quantum gravity: classical gravity is necessarily Markovian, while quantum gravity may permit non-Markovian information backflow.

In Paper 1b [2], we applied the τ framework to wavefunction collapse. A future paper (Paper 3) will investigate the weak-field limit at galactic scales.

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* akai@fawstudio.com

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